1.2

a、Performance v.s. Pipeling (流水線)

b、Dependability via Redundancy (由許多繩索共同支撐)

c、Performance via Prediction. (預先得到海象情報)

d、Make the Common Case Fast

e、Hierarchy of memories.

f、Performance via Parallelism

g、Design for Moore's Law

h、Use Abstraction to simplify Design.

1.5

P1: 3GHz, CPI = 1.5   P2: 2.5GHz, CPI = 1.0   P3: 4.0GHz, CPI = 2.2

a. $CPI = \frac{Cycles}{IC}$, $\text{Clock Rate} = \frac{Cycles}{\text{CPU Time}}$, $\frac{IC}{Cycles} \times \frac{Cycles}{\text{CPU Time}} = \frac{1}{CPI} \times \frac{Cycles}{\text{CPU Time}} = \frac{IC}{\text{CPU Time}}$

$IPS_{P1} = \frac{\overset{2}{\cancel{3}}\times 10^9}{\cancel{1.5}} = 2\times10^9$, $IPS_{P2} = \frac{2.5\times10^9}{1} = 2.5\times10^9$, $IPS_{P3} = \frac{4\times10^9}{2.2} \approx 1.8\times10^9$

b. Clock Cycles = CPU Time × Clock Rate

$Cycle_{P1} = 10 \times 3 \times \times 10^9 = 3\times10^{10}$ $IC_{P1} = \frac{\overset{2}{\cancel{3}}\times10^{10}}{\underset{1}{\cancel{1.5}}} = 2\times10^{10}$

$Cycle_{P2} = 10 \times 2.5\times10^9 = 2.5\times10^{10}$ $IC_{P2} = \frac{2.5\times10^{10}}{1.0} = 2.5\times10^{10}$

$Cycle_{P3} = 10 \times 4\times10^9 = 4\times10^{10}$ $IC_{P3} = \frac{4\times10^{10}}{22\times10^{-1}} \approx 1.818\times10^{10}$

c. $\text{Clock Rate} = \frac{\text{Clock Cycles}}{\text{CPU Time}} = \frac{IC \times CPI}{\text{CPU Time}}$

$\frac{1.2}{0.7} = \frac{12}{\cancel{10}} \times \frac{\cancel{10}}{7} = \frac{12}{7}$

$7 \overline{)30}$ 4.28

$\text{Clock Rate}_{P1} = 3\times10^9 \times \frac{12}{7} \approx 5.14\ GHz$

$\text{Clock Rate}_{P2} = 2.5\times10^9 \times \frac{12}{7} \approx 4.28\ GHz$

$\text{Clock Rate}_{P3} = 4.0\times10^9 \times \frac{12}{7} \approx 6.85\times10^9$

1.6

CMD \ CPU	P1 2.5GHz	P2 3.0GHz	
A	1	2	10^5
B	2	2	2×10^5
C	3	2	5×10^5
D	3	2	2×10^5

a.

$$CPI_{P1} = \frac{(1\times10^5 + 4\times10^5 + 15\times10^5 + 6\times10^5)}{10^6} = 2.6$$

$$CPI_{P_2} = \frac{2\times10^6}{10^6} = 2 \cancel{\#}$$

b.

$$P1 : 10^5 + 4\times10^5 + 15\times10^5 + 6\times10^5 = 2.6\times10^6$$

$$P2 : 2\times10^6$$

1.7 $CPI = \frac{Cycles}{IC} = \frac{CPU\ Time \times Clock\ Rate}{IC}$

Clock Cycles Time = 1ns
Clock Rate = 1.0 GHz

a. $IC_A : 10^9$, 1.1s $\quad CPI_A = \frac{1.1\times\cancel{10^9}}{\cancel{10^9}} = 1.1$

$IC_B : 1.2\times10^9$ 1.5 $\quad CPI_B = \frac{1.5\times\cancel{10^9}}{1.2\times\cancel{10^9}} = 1.25$

b. $CPU\ Time = \frac{IC \times CPI}{Clock\ Rate}$ $(Time_A = Time_B)$

$$\frac{1.2\times\cancel{10^9}\times1.25}{1.1\times\cancel{10^9}} \approx 1.37\text{(倍)}$$

c.

$IC_C : 6\times10^8$, $CPI_C = 1.1$

$$\frac{Time_A}{Time_C} = \frac{10^{\cancel{9}}\times1.1}{6\times\cancel{10^8}\times1.1} = \frac{11}{6.6} \approx 1.67 \qquad \frac{Time_B}{Time_C} = \frac{1.5\times10^{\cancel{9}}}{6\times10^{\cancel{8}}\times1.1} = \frac{15}{6.6} \approx 2.27$$

1.9

1.9-1

CPUs	Arit. CMD (CPI = 1)	L/S CMD (CPI = 12)	Branch CMD (CPI = 5)	Cycles	Ex. Time	Speed Up
1	2.56×10^9	1.28×10^9	2.56×10^8	1.92×10^{10}	9.6	1
2	1.83×10^9	9.14×10^8	2.56×10^8	1.41×10^{10}	7.05	1.36
4	9.12×10^8	4.57×10^8	2.56×10^8	7.68×10^9	3.84	2.5
8	4.57×10^8	2.29×10^8	2.56×10^8	4.48×10^9	2.24	4.29

1.9-2

CPUs	Ex. Time
1	10.88
2	7.95
4	4.29
8	2.47

1.9-3

$$\frac{2.56\times10^9 + x\times1.28\times10^9 + 5\times2.56\times10^8}{2.0\times10^9} = 3.84$$

$$\frac{2.56\cancel{\times10^9} + 1.28x\cancel{\times10^9} + 1.28\cancel{\times10^9}}{2.0\cancel{\times10^9}} = 3.84$$

$3.84 + 1.28x = 7.68$, $1.28x = 3.84$, $x = 3$ #

1-11

1-11.1 $\text{Clock Rate} = \dfrac{1}{0.333\times10^{-9}} \approx 3.003\times10^9$ (3.003 GHz)

$\text{CPU Time} = \dfrac{\text{IC}\times\text{CPI}}{\text{Clock Rate}}$, $\text{CPI} = \dfrac{\text{Time}\times\text{Clock Rate}}{\text{IC}} = \dfrac{7.5\times3.003\times10^{4}}{2.389\times10^{12}} \approx 0.94$

1-11.2

$\dfrac{9650}{750} \approx 12.86$ #

1-11.3

$750\times0.1 = 75$ (多了75秒)

1-11.4

$1.1\times1.05 = 1.155$

$750\times0.155 = 116.25$

(增加了116.25秒)

1-11.5

$\dfrac{9650}{866.25} \approx 11.14$

$12.86 - 11.14 = 1.72$

1-11.6

$$CPI = \frac{7 \times 4 \times 10^{11}}{2.389 \times 10^{12} \times 8.5 \times 10^{-1}} = \frac{2.8 \times 10^{12}}{2.0307 \times 10^{12}} \approx 1.37 \#$$

1-11.7

$$CPI_{\uparrow} = \frac{1.37}{0.94} \approx 1.457 \ (+46\%) \qquad Rate_{\uparrow} = \frac{4 \times 10^9}{3 \times 10^9} \approx 1.33 \ (33\%)$$

雖CMD數減少了15%，但CPU Time降低的幅度更少；所以兩者不相近

1-11.8

$$\frac{700}{750} \approx 0.93 \ (\text{減少了}6.7\%)$$

1-11.9

1-11.9

$$IC = \frac{T \times Rate}{CPI} = \frac{960 \times 9 \times 10^{-1} \times 4 \times 10^9}{1.61} = \frac{3.456 \times 10^{12}}{1.61} \approx 2.1466 \times 10^{12}$$

1-11.10

$$Rate = \frac{IC \times CPI}{T} = \frac{4 \times 10^9}{9 \times 10^{-1}} = 0.44 \times 10^{10} = 4.4 \times 10^9 \ (4.4\ GHz)$$

1-11.11

$$Rate = \frac{IC \times 0.85 CPI}{0.8 T} \approx \frac{17}{16} \times \frac{IC \times CPI}{T} \quad 3 \times 10^9 \times \frac{17}{16} \approx 3.18 \times 10^9$$

(3.18 GHz

1-12

1-12.1

$$T_1 = \frac{5\times\cancel{10^9}\times 0.9}{4\times\cancel{10^9}} = 1.125 \qquad T_2 = \frac{\cancel{10^9}\times 0.75}{3\times\cancel{10^9}} = 0.25$$

$Rate_1 > Rate_2$, but $Performance_1 < Performance_2$

1-12.2

$$T_1 = \frac{\cancel{10^9}\times 0.9}{4\times\cancel{10^9}} = 0.225 \quad , \quad IC_2 = \frac{0.225\times 3\times 10^9}{0.75} = 9\times 10^8$$

1-12.3

$$MIPS_1 = \frac{4\times 10^{\cancel{9}\,4}}{9\times\cancel{10^5}} \approx 4.44\times 10^3 \quad MIPS_2 = \frac{3\times 10^{\cancel{9}\,5}}{75\times\cancel{10^4}} = 4\times 10^3$$

$MIPS_1 > MIPS_2$, but $Performance_1 < Performance_2$

1-12.4

$$MFLOPS = \frac{\text{Float IC}}{Ex\,T\times 10^6}$$

$$MFLOPS_1 = \frac{5\times 10^{\cancel{9}\,\cancel{8}\,2}\times\cancel{4\times 10^{-1}}}{1.125\times\cancel{10^6}} = \frac{2\times 10^3}{1.125} \approx 1.78\times 10^3$$

$$MFLOPS_2 = \frac{1.0\times 10^{\cancel{9}\,\cancel{8}\,3}\times\cancel{4\times 10^{-1}}}{2.5\ \cancel{\times 10^5}} = \frac{4\times 10^3}{2.5} = 1.6\times 10^3$$

$MFLOPS_1 > MFLOPS_2$, but $Performance_1 < Performance_2$